

Supplementary results for first hospitalisation counts before and during the COVID pandemic

The tables distinguish disjoint descriptive periods from overlapping model fits. Confidence intervals are nominal pointwise 95% normal-reference Wald intervals for exploratory associations; no later-era effect is inferred by subtracting nested coefficients.

Annual recorded counts

Supplementary Table S1. Annual recorded first hospitalisations after coronary heart disease (CHD) or heart failure (HF) diagnosis, 2013–2023. Each total is the sum of twelve supplied monthly counts. Admission cause and eligible person-time were unavailable; no incidence rate or unique combined patient count is implied.

Year	CHD first hospitalisations	HF first hospitalisations
2013	23,830	4,336
2014	17,760	3,675
2015	15,411	2,934
2016	14,510	2,734
2017	13,962	2,620
2018	12,616	2,527
2019	12,396	2,344
2020	10,237	1,964
2021	12,035	2,275
2022	11,076	1,976
2023	12,323	2,296

Descriptive periods

Supplementary Table S2. Monthly mean recorded counts in non overlapping calendar periods. Coronary heart disease (CHD) and heart failure (HF) arithmetic means from aggregate period summaries.

Period	Months	CHD monthly mean	HF monthly mean
2013–2019	84	1315.3	252.0
2020–2022	36	926.3	172.6
2023	12	1026.9	191.3

Complete baseline panel

Supplementary Table S3. All twelve baseline weather models, January 2013–December 2023. CHD denotes coronary heart disease; HF denotes heart failure. Days-in-month offset, calendar-month indicators, four-degree-of-freedom natural time spline; Newey–West lag-six intervals. BH q-values cover these twelve two-sided Wald tests of zero exposure coefficient only; interval endpoints are not multiplicity-adjusted. The window spans 132 calendar months, not 132 independent replicates.

Outcome	Exposure	Count ratio and 95% CI	p	BH q
CHD	Mean temperature / 1°C	0.993 (0.976–1.010)	0.424	0.726
CHD	Mean maximum / 1°C	0.993 (0.979–1.007)	0.321	0.642
CHD	Mean minimum / 1°C	0.994 (0.977–1.012)	0.509	0.764
CHD	Hot nights / 5 days	1.022 (1.002–1.042)	0.032	0.192
CHD	Very hot days / 5 days	0.999 (0.974–1.025)	0.962	0.962
CHD	Cold days / 5 days	0.995 (0.949–1.043)	0.823	0.900
HF	Mean temperature / 1°C	0.974 (0.947–1.002)	0.069	0.207
HF	Mean maximum / 1°C	0.981 (0.958–1.005)	0.113	0.272
HF	Mean minimum / 1°C	0.973 (0.947–1.000)	0.050	0.202
HF	Hot nights / 5 days	1.003 (0.976–1.031)	0.825	0.900
HF	Very hot days / 5 days	0.995 (0.963–1.028)	0.764	0.900
HF	Cold days / 5 days	1.073 (1.006–1.144)	0.031	0.192

The HF mean-minimum-temperature lag-six interval includes one on the unrounded scale; its upper bound is 1.0000496637 (Wald $p = 0.0504$). Rounding to 1.000 in Table S3 or 1.0000 in Table S6 does not change that interpretation.

The nested pre-2020 continuous-temperature fits were CHD mean temperature 0.980 (0.970–0.990), mean maximum 0.982 (0.974–0.991), and mean minimum 0.982 (0.972–0.992); HF corresponding fits were 0.945 (0.927–0.963), 0.958 (0.939–0.977), 0.944 (0.927–0.961). These estimates are not multiplicity-adjusted or a later-era contrast.

Overlapping analysis windows

Supplementary Table S4. Official-day associations in nested (2013–2019) and full (2013–2023) windows. CHD denotes coronary heart disease; HF denotes heart failure. Ratios per five additional days with Newey–West lag-six 95% intervals. The pre-2020 window is contained within the full series, with a separately reconstructed spline basis. These are dependent sensitivity fits, not independent pre/post estimates.

Outcome	Exposure	Pre 2020 84 months	Full 132 months
CHD	Hot nights / 5 days	1.011 (0.991–1.032)	1.022 (1.002–1.042)
CHD	Very hot days / 5 days	0.997 (0.982–1.012)	0.999 (0.974–1.025)
CHD	Cold days / 5 days	1.036 (1.007–1.067)	0.995 (0.949–1.043)
HF	Hot nights / 5 days	0.965 (0.937–0.994)	1.003 (0.976–1.031)
HF	Very hot days / 5 days	0.990 (0.960–1.021)	0.995 (0.963–1.028)
HF	Cold days / 5 days	1.113 (1.053–1.176)	1.073 (1.006–1.144)

Calendar specification sensitivity

Supplementary Table S5. Complete official day panel under alternative calendar specifications. Coronary heart disease (CHD) and heart failure (HF) ratios per five days with Newey–West lag-six intervals. The full series is January 2013–December 2023. Baseline and nested values are in Supplementary Table S4. Year indicators replace the spline. COVID-phase intercepts supplement the baseline in the full series; they are not a third window. Omission checks shorten the 132-calendar-month window to 120 or 108 months. These window lengths are not independently verified fitted-observation counts. These sensitivity estimates have no multiplicity adjustment.

CHD

Specification	Hot nights	Very hot days	Cold days
Time spline 3 df	1.011 (0.990–1.032)	0.993 (0.967–1.020)	0.989 (0.934–1.046)
Time spline 6 df	1.024 (1.005–1.044)	0.999 (0.975–1.023)	0.999 (0.952–1.047)
Time spline 8 df	1.023 (1.005–1.042)	0.997 (0.974–1.021)	0.997 (0.950–1.047)
Year indicators	1.025 (1.000–1.050)	1.006 (0.983–1.029)	0.994 (0.939–1.052)
Omit first 12 months	1.020 (1.000–1.040)	0.995 (0.969–1.022)	1.008 (0.954–1.065)
Omit first 24 months	1.017 (0.999–1.036)	0.993 (0.966–1.020)	0.989 (0.930–1.053)
COVID phase intercepts	1.013 (0.994–1.033)	0.993 (0.973–1.013)	0.991 (0.943–1.041)

HF

Specification	Hot nights	Very hot days	Cold days
Time spline 3 df	0.993 (0.963–1.025)	0.989 (0.956–1.024)	1.066 (1.005–1.131)
Time spline 6 df	1.004 (0.977–1.032)	0.995 (0.963–1.028)	1.072 (1.005–1.143)
Time spline 8 df	1.002 (0.978–1.028)	0.993 (0.963–1.024)	1.062 (0.994–1.135)
Year indicators	1.004 (0.982–1.026)	0.997 (0.971–1.024)	1.061 (0.997–1.129)
Omit first 12 months	1.007 (0.977–1.038)	0.995 (0.963–1.029)	1.067 (0.995–1.143)
Omit first 24 months	1.003 (0.974–1.034)	0.988 (0.957–1.021)	1.043 (0.965–1.127)
COVID phase intercepts	0.994 (0.972–1.016)	0.990 (0.963–1.017)	1.074 (1.006–1.145)

Uncertainty constructions

Supplementary Table S6. Uncertainty intervals for all twelve baseline models, January 2013–December 2023. CHD denotes coronary heart disease; HF denotes heart failure. All four constructions use the same fitted coefficient. Lag-six q-values appear in Table S3 and do not apply to the other interval constructions. Four decimal places are retained for the near-null CHD hot-night lag-three bounds.

CHD

Exposure	Model CI	HC1 CI	NW3 CI	NW6 CI
Mean temperature / 1°C	0.9772–1.0092	0.9745–1.0120	0.9759–1.0105	0.9762–1.0102
Mean maximum / 1°C	0.9788–1.0073	0.9764–1.0098	0.9784–1.0077	0.9791–1.0069
Mean minimum / 1°C	0.9788–1.0098	0.9763–1.0124	0.9768–1.0119	0.9771–1.0116
Hot nights / 5 days	0.9950–1.0494	0.9973–1.0469	1.0003–1.0439	1.0019–1.0422
Very hot days / 5 days	0.9741–1.0254	0.9747–1.0247	0.9735–1.0260	0.9744–1.0251
Cold days / 5 days	0.9539–1.0371	0.9555–1.0353	0.9495–1.0419	0.9488–1.0427

HF

Exposure	Model CI	HC1 CI	NW3 CI	NW6 CI
Mean temperature / 1°C	0.9561–0.9929	0.9494–0.9999	0.9487–1.0006	0.9474–1.0020
Mean maximum / 1°C	0.9645–0.9981	0.9584–1.0044	0.9593–1.0035	0.9583–1.0046
Mean minimum / 1°C	0.9556–0.9909	0.9500–0.9967	0.9483–0.9986	0.9469–1.0000
Hot nights / 5 days	0.9702–1.0372	0.9706–1.0367	0.9739–1.0332	0.9759–1.0311
Very hot days / 5 days	0.9637–1.0274	0.9629–1.0282	0.9637–1.0273	0.9630–1.0281
Cold days / 5 days	1.0227–1.1253	1.0113–1.1381	1.0066–1.1434	1.0065–1.1435

Released baseline diagnostics

Supplementary Table S7. Diagnostics from all twelve baseline models, January 2013–December 2023. CHD denotes coronary heart disease; HF denotes heart failure. Theta is the negative-binomial dispersion parameter. Pearson dispersion, lag-one Pearson residual autocorrelation and lag-six Ljung–Box p-values describe the released fits; no new diagnostic test was performed. Small Ljung–Box p-values indicate residual dependence; they do not establish interval calibration. Sandwich covariance depends on regression-score autocovariances, so positive residual autocorrelation alone does not require HAC intervals to widen. All released baseline convergence flags were true.

Outcome	Exposure	Theta	Pearson dispersion	Residual ACF1	Ljung–Box p lag6
CHD	Mean temperature	160.456	1.114	0.534	1.24e-09
CHD	Mean maximum	160.760	1.114	0.533	1.36e-09
CHD	Mean minimum	160.190	1.114	0.532	1.48e-09
CHD	Hot nights	163.009	1.116	0.508	1.17e-08
CHD	Cold days	159.450	1.114	0.528	2.24e-09
CHD	Very hot days	159.359	1.114	0.529	2.05e-09
HF	Mean temperature	186.459	1.149	0.139	0.469
HF	Mean maximum	178.627	1.147	0.150	0.397
HF	Mean minimum	190.800	1.151	0.131	0.514
HF	Hot nights	164.480	1.139	0.174	0.364
HF	Cold days	188.039	1.149	0.146	0.356
HF	Very hot days	164.636	1.140	0.179	0.319

Interpretation and source binding

The pre-2020 window contains 84 of the 132 full-window calendar months, with a recalculated spline basis. Released metadata record window length rather than the number of observations used by each fitted model. Complete-case handling was not independently documented; this does not establish that any observations were missing. No independent 2020–2023 coefficient, interaction test or weather standardisation is reported. General-population rates in older aggregate files are omitted because the population is not the eligible first-event risk set.

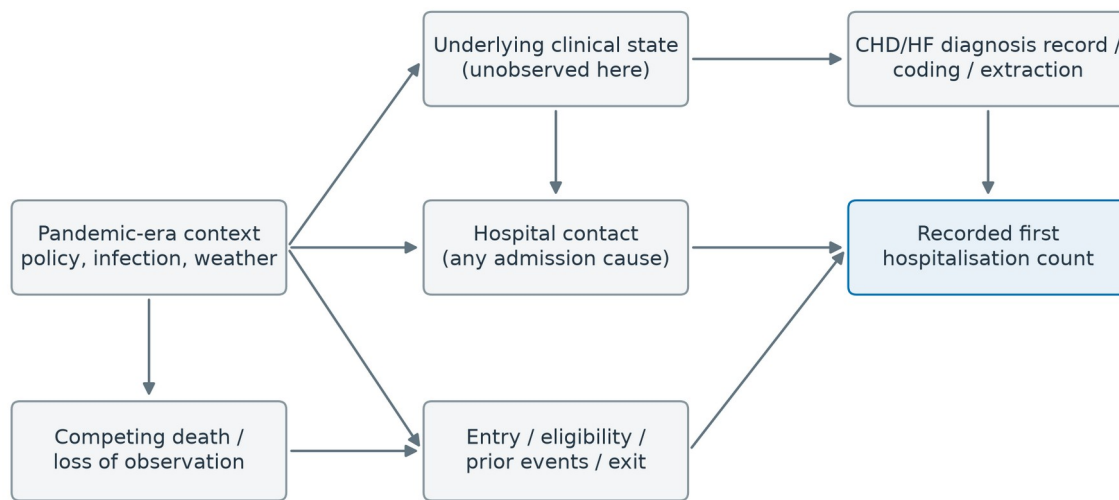
Tables draw from the repository files `outputs/tables/cvd_descriptive_annual_totals.csv`, `cvd_descriptive_covid_era_means.csv`, `cvd_core_robust_estimates.csv` and

cvd_trend_depletion_sensitivity.csv. Baseline diagnostics are in cvd_core_model_fit.csv. All health values have HA_APPROVED_AGGREGATE provenance; no monthly governed files were read to produce this revision.

Conceptual observation process

Supplementary Figure S1. Observation processes underlying a recorded first stay. CHD denotes coronary heart disease; HF denotes heart failure. Conceptual diagram of how underlying clinical state, diagnosis recording, care contact, eligibility and competing death can affect the supplied count. Arrows specify plausible relationships with unknown signs; they are not identified effects or a complete causal adjustment graph. Admission cause, eligible person-time and linked clinical measurements were unavailable. No numerical cohort flow is inferred.

Conceptual observation process



Candidate pathways only. Counted stays follow a diagnosis record; their admission cause is unavailable.

Visual comparison of nested and full windows

Supplementary Figure S2. Official day associations in overlapping analysis windows. CHD denotes coronary heart disease; HF denotes heart failure. Points and Newey–West lag-six 95% confidence intervals for 2013–2019 and 2013–2023 reproduce Supplementary Table S4. The pre-2020 84 months are contained within the full 132 months, and the spline basis is recalculated in each fit. The paired display is a sensitivity comparison, not independent pre/post estimates or a test of a change. Ratios are per five additional official days, without a consecutive-day requirement.

